



Financial Model

Bottom-up financial projection across three scenarios. All figures USD. Projections are illustrative. Stress-test against actual pipeline data as sales begin.

Decision Graph Engine

AI-Native MBSE

SysML v2

Regulated Industries

01 Market Sizing

TAM (2026)

\$4.5B

15–16% CAGR → \$9–10B by 2031

SAM

\$1.8B

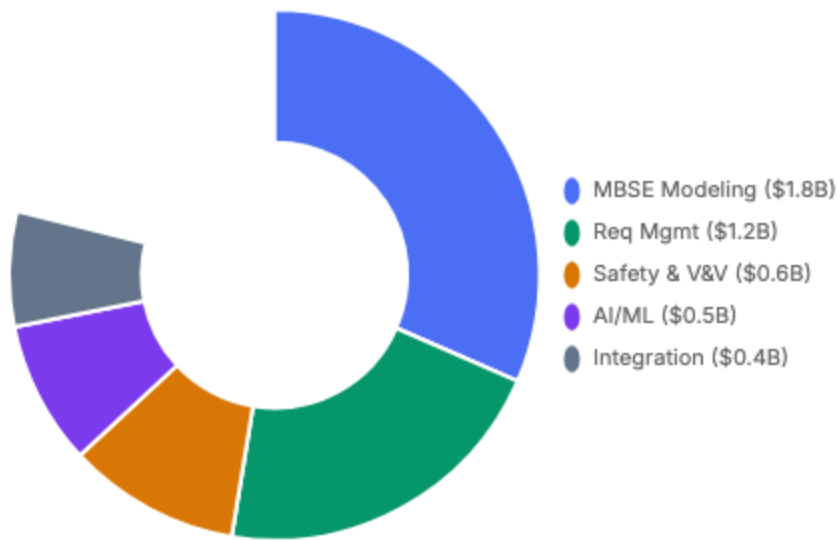
English-speaking, transition-ready

Y5 SOM TARGET

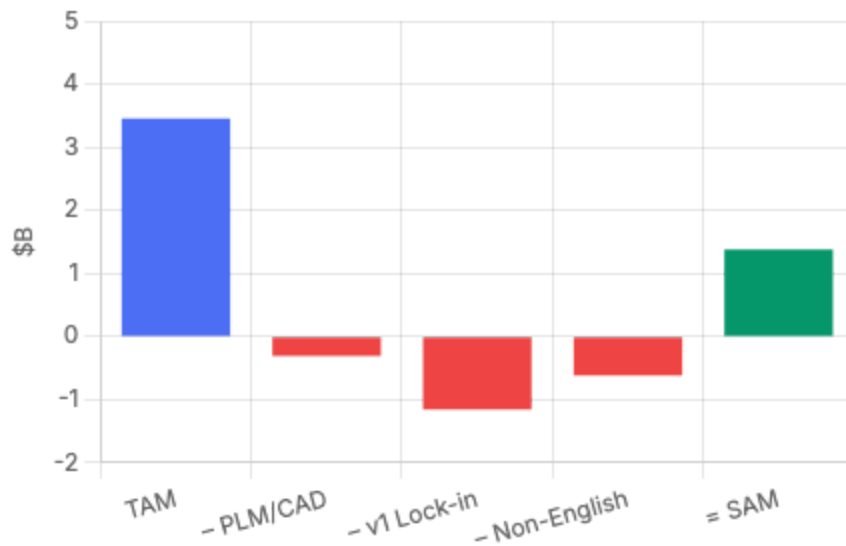
\$30–70M

Category leadership positioning

TAM Breakdown by Segment



TAM → SAM Waterfall



SOM Targets by Year

Year	Target SOM	Assumptions
Y1	\$0.5–1.5M	3–8 design partners, subsidized pricing
Y2	\$2–5M	10–25 paying customers, first enterprise
Y3	\$6–15M	Expansion seats, first defense/gov
Y4	\$15–35M	Scaling enterprise, international begins

Year	Target SOM	Assumptions
Y5	\$30–70M	Category leadership, platform revenue

02 Revenue Model & Pricing

TEAM

\$100/seat/mo

5–20 seats · SysML v2 · collab · req mgmt · basic traceability

PROFESSIONAL

\$200/seat/mo

20–100 seats · + safety analysis · INCOSE · ambient ML · API

ENTERPRISE

\$300/seat/mo

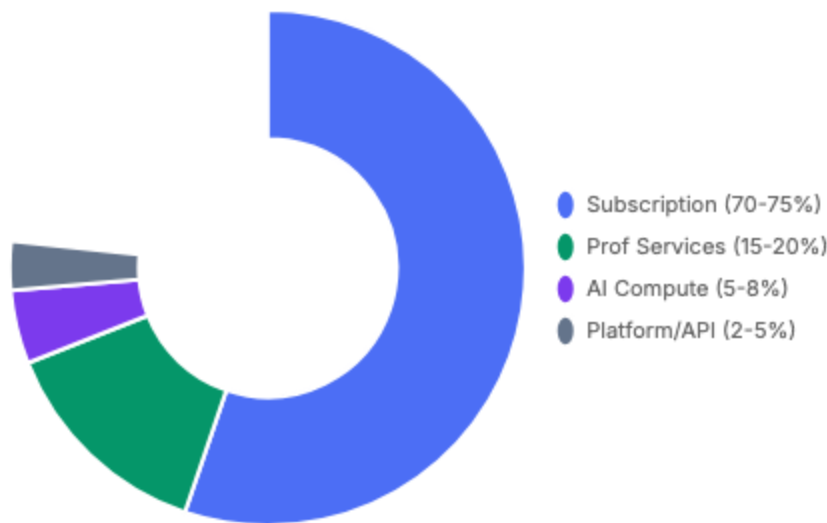
100+ seats · + on-prem · SSO · audit logs · SLA

DEFENSE / GOV

\$450/seat/mo

50+ seats · + air-gap · local AI · IL4/IL5 · FedRAMP

Revenue Mix at Scale

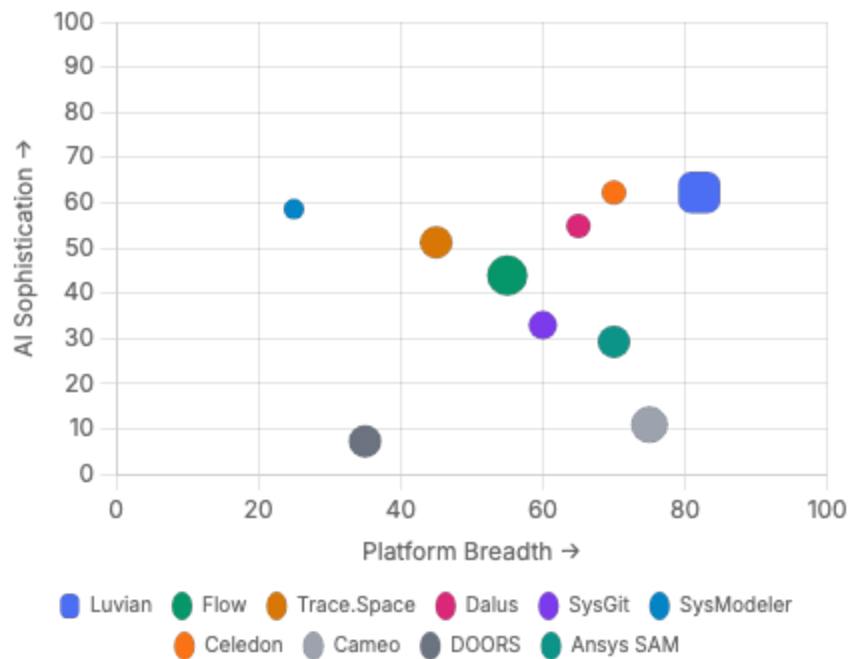


Deal Size by Segment

Segment	Min Seats	Typical ACV	Max ACV
Startup	5	\$15K	\$50K
Mid-market	20	\$100K	\$250K
Enterprise	100	\$500K	\$1.5M
Defense (program)	50	\$400K	\$2M
Defense (enterprise)	500+	\$5M	\$15M+

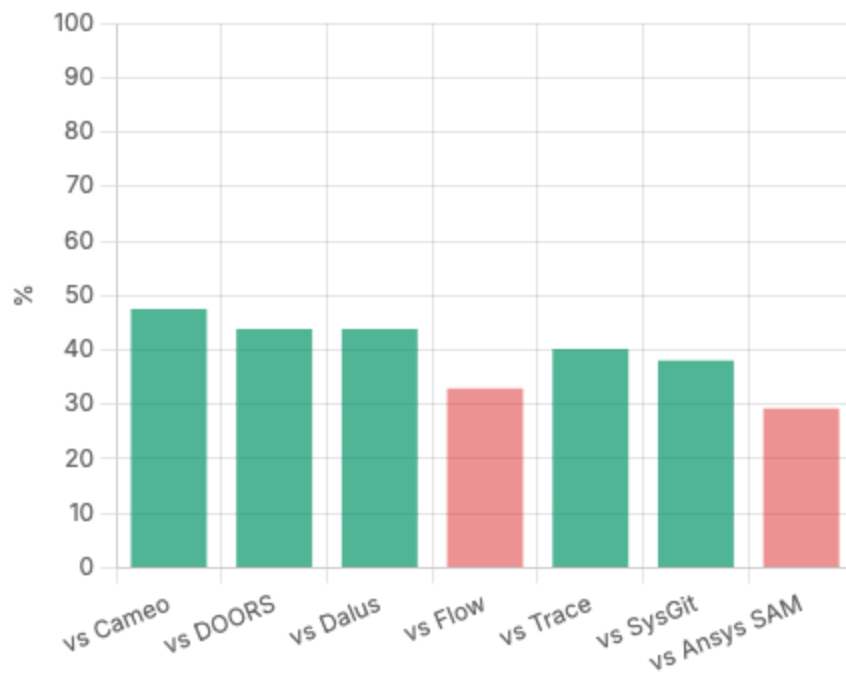
03 Competitive Landscape

Competitive Positioning (Scope × AI)



X = Platform breadth · Y = AI sophistication · Size = funding

Head-to-Head Win Rate (est.)



Tier 1 - Well-Funded Direct Competitors (>\$10M raised)

Company	Funding	Price	Core Strength	Luvian Edge
Flow Engineering	\$35.8M (Sequoia)	\$150–\$300/mo	Req mgmt for iterative HW; 1K+ Rivian engs	Safety + SysML v2 modeling (Flow has neither)
Trace.Space	~\$11.5M	Custom	AI-native req + traceability; SOC 2 + ISO 27001	Integrated modeling + safety (Trace is req-only)
Ansys SAM	Corporate	~\$2K–\$10K/yr	SysML v2 full spec; Ansys simulation ecosystem	Independent; AI-native; 10× lower cost

Tier 2 - Early-Stage Direct Competitors (<\$10M raised)

Company	Funding	Price	Core Strength	Luvian Edge
Dalus	~\$500K (YC W25)	\$99–\$249/mo	AI copilot; SOC 2 Type 2; GovCloud	Deeper INCOSE + ambient ML + on-prem air-gap
SysGit	\$4.1M + \$2.2M SBIR	Custom	Git-based SE; defense-native; SysML v2	Visual canvas; AI suggestions; broader verticals

Company	Funding	Price	Core Strength	Luvian Edge
Systellar	Pre-seed (ESA BIC)	N/A	SysML v2; CI/CD pipelines for models	US market; deeper AI; more mature
SysModeler.ai	Unfunded	Free (credits)	AI-first SysML diagram gen	Full platform vs. single feature
Celedon Davinci	SBIR Phase II	Free (beta)	Multi-agent AI engineering; NASA use	Production arch; real collab; standards compliance
RoboFication	Bootstrapped	N/A	AI for safety analysis; Ford alumni	Broader scope (modeling+req+safety)

Tier 3 - Incumbents & Adjacent

Company	Type	Typical Price	Luvian Position
Cameo / MagicDraw	Incumbent MBSE	\$5K-\$15K/yr	3-10× cheaper; AI; web; collab
IBM DOORS Next	Incumbent req mgmt	\$5K-\$20K/yr	3-15× cheaper; integrated modeling
Jama Connect	Req management	\$5K-\$12K/yr	SysML v2 native; safety built-in
Intercax Syndeia	Digital thread	Custom	Syndeia connects; Luvian replaces
UST	Eng services (\$5B+)	Consulting	Potential channel partner

04 Customer Acquisition

Sales Cycle & Win Rate

Segment	Cycle	Win Rate	CAC
Startup	2-4 wks	25%	\$3-5K

Segment	Cycle	Win Rate	CAC
Mid-market	2–4 mo	15–20%	\$15–30K
Enterprise	4–9 mo	10–15%	\$50–100K
Defense/Gov	9–18 mo	8–12%	\$100–250K

Net Revenue Retention Targets



05 Cost Structure

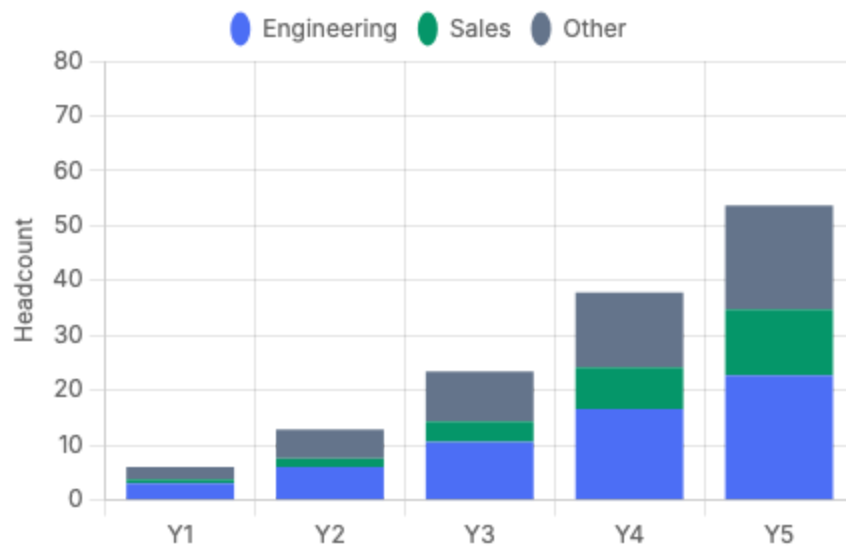
Y1 Cost Breakdown (\$1.72M)



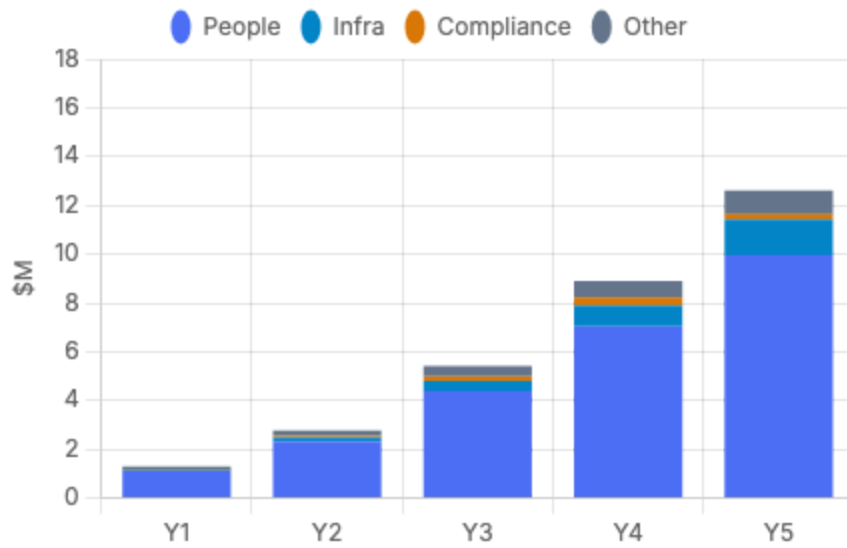
Y5 Cost Breakdown (\$16.77M)



Headcount Growth by Function

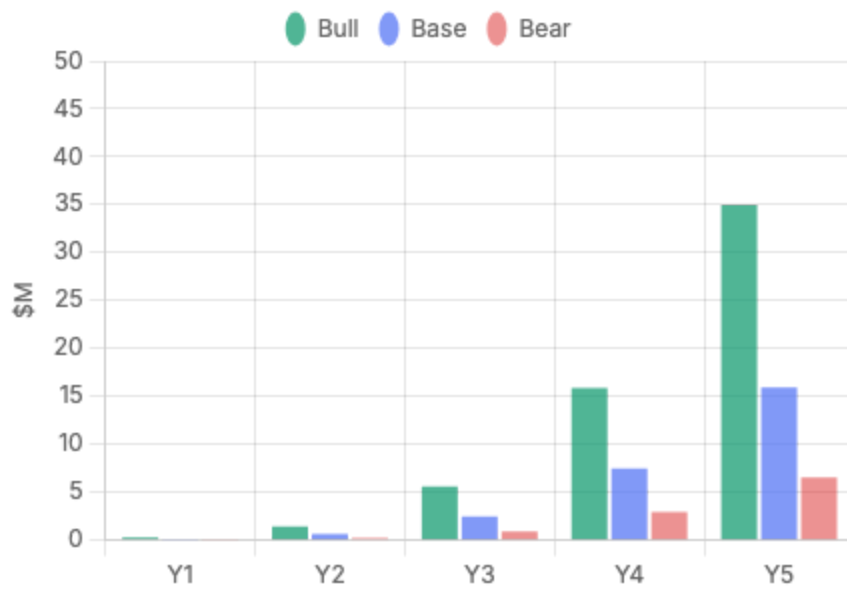


Total Cost by Category (\$M)

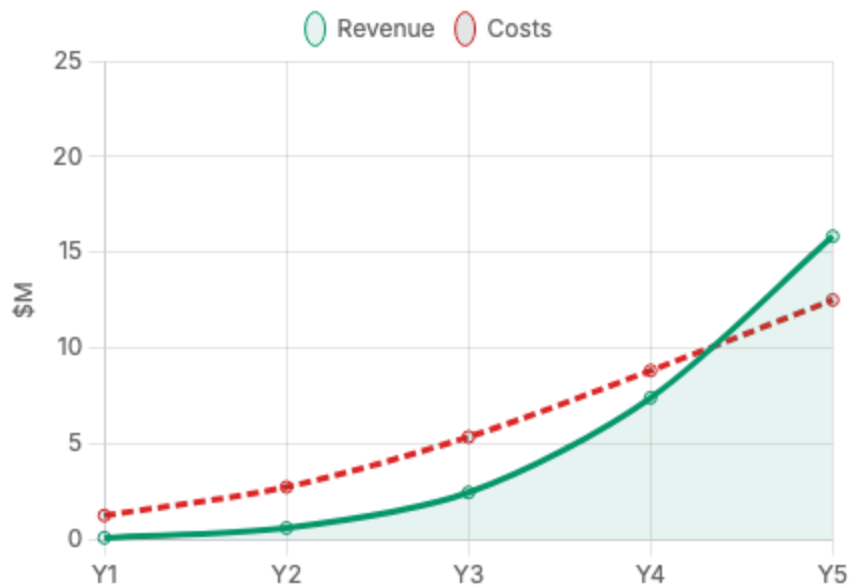


06 Five-Year P&L Projections

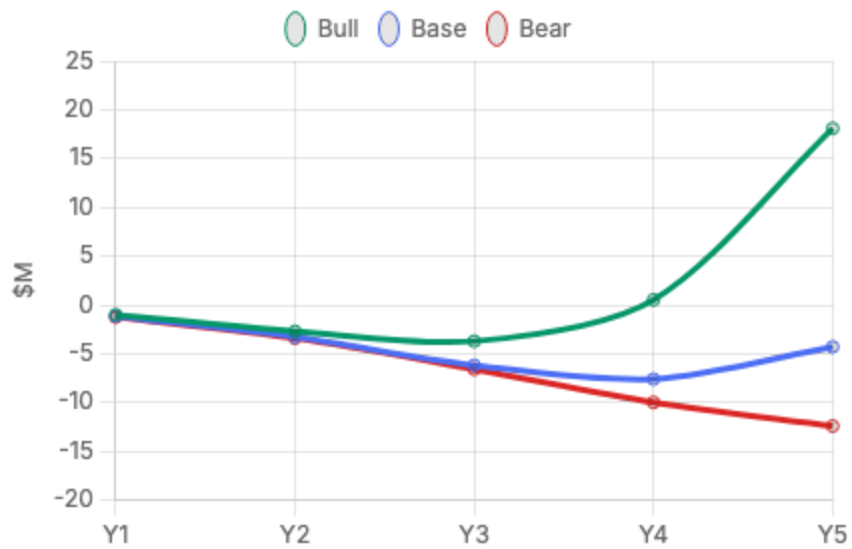
Revenue by Scenario (\$M)



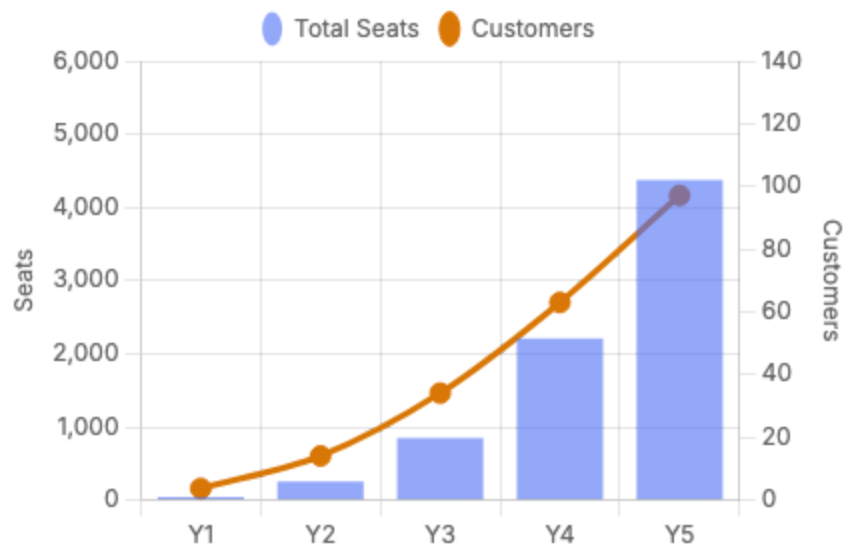
Base Case: Revenue vs. Costs (\$M)



Cumulative Burn by Scenario (\$M)



Customer & Seat Growth (Base)

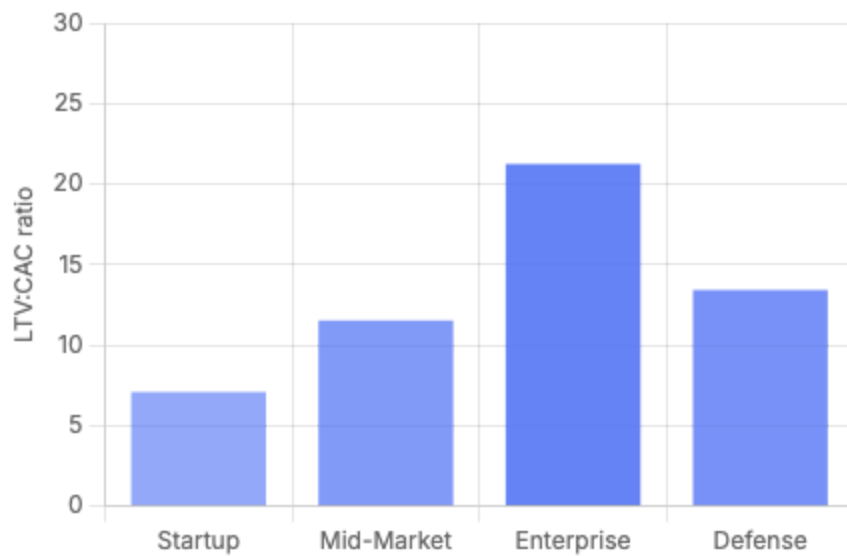


Base Case P&L Detail

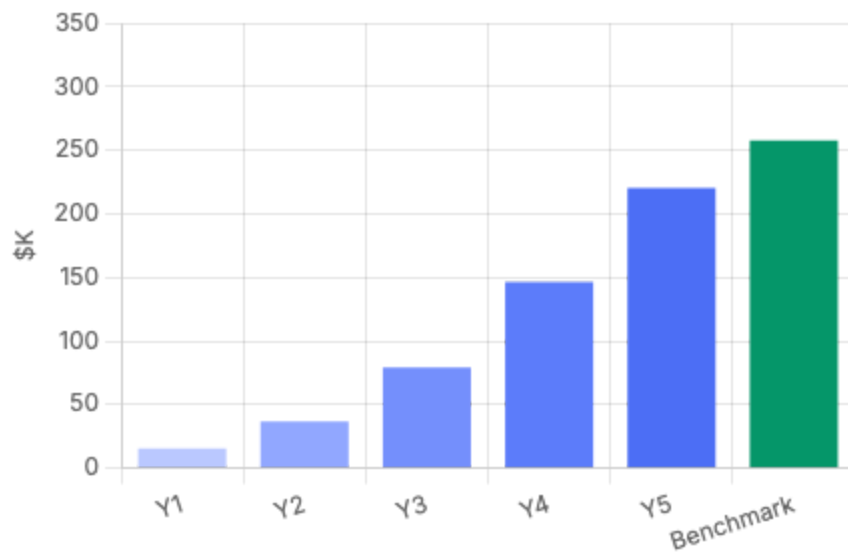
	Y1	Y2	Y3	Y4	Y5
New customers	5	15	30	45	55
Total customers	5	19	46	85	131
Total seats	60	342	1,150	2,975	5,895
Total Revenue	\$165K	\$846K	\$3.33M	\$9.94M	\$21.21M
Total Costs	\$1.72M	\$3.71M	\$7.21M	\$11.84M	\$16.77M
Net Income	(\$1.56M)	(\$2.86M)	(\$3.88M)	(\$1.90M)	\$4.44M

07 Unit Economics

LTV:CAC by Segment (target >3x)



Revenue per Employee (\$K)



Gross Margin by Revenue Stream

Stream	COGS	Gross Margin
Subscription (SaaS)	Hosting, DB, support	82-85%
Subscription (on-prem)	Support, packaging	85-88%
Professional services	SE labor, travel	40-55%
AI compute	LLM API, hosting	55-65%

Stream

COGS

Gross Margin

Blended (at scale)

75-80%

08 Funding Model

SEED

\$7-10M

Y1 Q1 · 5 design partners · SOC 2

SERIES A

\$20-30M

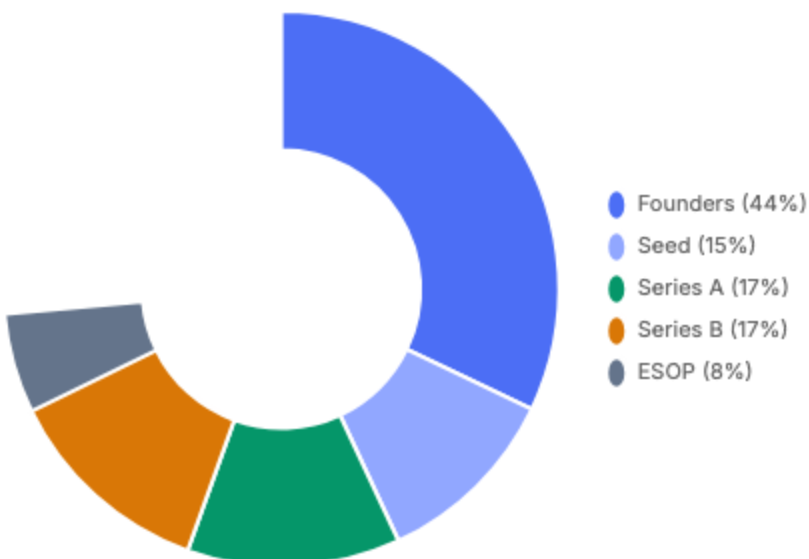
Y2 · \$1M+ ARR gate · FedRAMP start

SERIES B

\$50-80M

Y3-Y4 · \$5M+ ARR · Defense contracts

Post-Series B Ownership



Peak Cash Need by Scenario

Scenario	Peak Cash	Raise Needed	Breakeven
Bull	\$5.0M	\$6–8M	Month 38
Base	\$10.2M	\$12–15M	Month 50
Bear	\$16.7M+	\$18–22M	Not in 5 yrs

09 Key Metrics Dashboard

Metric	Y1	Y3	Y5
ARR	\$90K+	\$2.5M+	\$16.5M+
NRR	100%	120%	130%
Customers	5	46	131
Avg seats/customer	12	25	45
Logo churn (annual)	<15%	<10%	<8%
Burn multiple	N/A	<3×	<1.5×
Gross margin	80%	78%	80%

10 Sensitivity Analysis

Y5 Revenue Impact of ±20% Change (\$M)

11 Competitive Reality Check

Pricing Is Not a Differentiator

Every startup is \$100–\$300/mo. "10× cheaper than Cameo" is table stakes. Dalus is \$99/mo. Flow is \$150/mo.

Compliance Is Behind

Dalus has SOC 2 Type 2 + GovCloud. Trace.Space has SOC 2 + ISO 27001. Flow has ITAR GovCloud. Luvian has none.

Flow Is the Real Threat

\$35.8M from Sequoia. 1,000+ Rivian engineers. But: no MBSE, no SysML v2, no safety analysis.

Luvian's Defensible Position

No competitor covers modeling + requirements + safety + V&V in one auditable graph. This is the moat.

12 Strategic Revenue Levers

Y1: Close the Compliance Gap

- SOC 2 Type 1 immediately (\$50–80K, month 1)
- 5 design partners in regulated industries

- Launch free tier (1 project, 3 users, 100 elements)
- File provisional patents

Y2: Win the Regulated Vertical

- SOC 2 Type 2 + ISO 27001 by mid-Y2
- First AE (automotive/aerospace background)
- "Why not Flow?" battle card for every deal
- INCOSE/OMG conference presence

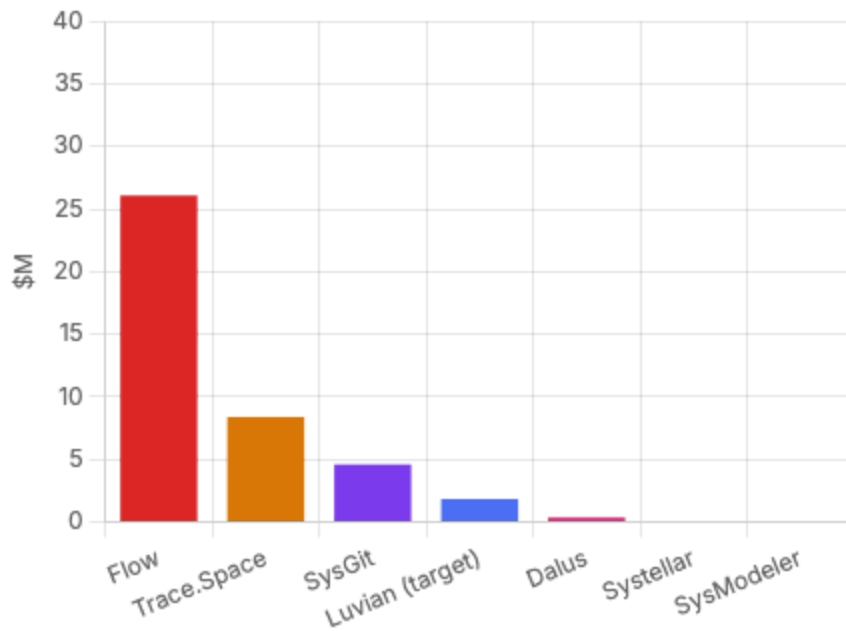
Y3: Standard for Safety-Critical

- Channel partners (Booz Allen, Leidos, SAIC)
- FedRAMP Li-SaaS certification
- Nominal partnership for verification digital thread
- Content engine: compliance evidence case studies

Y4-5: Platform & Ecosystem

- IL4/IL5 opens DoD (\$2M-\$15M ACV)
- Platform API revenue + ecosystem lock-in
- International: UK, Germany, France
- Acquire complementary niche tools

13 Competitive Funding Comparison



Flow has 5x more capital than Luvian's target seed. Series A of \$20–30M positions us to compete at scale.

14 Valuation Benchmarks

Private Competitor Valuations

Company	Last Round	Est. Post-Money
Flow Engineering	Series A (Oct 25)	~\$120–180M
Trace.Space	Seed (Jul 25)	~\$40–60M
SysGit	Seed (Sep 23)	~\$20–30M
Dalus	YC Seed (Apr 25)	~\$5–8M

Public Company Multiples

Company	Revenue	Multiple
Palantir	\$2.8B	25x+
Ansys	\$2.3B	15x

Company	Revenue	Multiple
PTC	\$2.1B	10×
GitLab	\$700M	12×

15 Model Assumptions Register

#	Assumption	Confidence	Validation
1	MBSE market grows 15%+ CAGR through 2031	High	Analyst reports
2	SysML v2 transition creates replacement window	Medium	OMG timeline; interviews
12	No well-funded competitor → Competitors exist and are funded	Revised	Flow \$36M, Trace \$11.5M, etc.
13	Integrated scope (model+req+safety+V&V) is defensible	Med-High	No competitor covers all four
15	Compliance (SOC 2, FedRAMP) is table stakes	Revised ↑	Dalus, Trace already certified

Luvian Confidential - NDA Restrictions. This model should be updated quarterly as revenue data, pipeline metrics, and cost actuals become available.

Source: docs/private/FINANCIAL_MODEL.md · Last updated 2026-04-07